# K-Means

## K-Mean Clustering

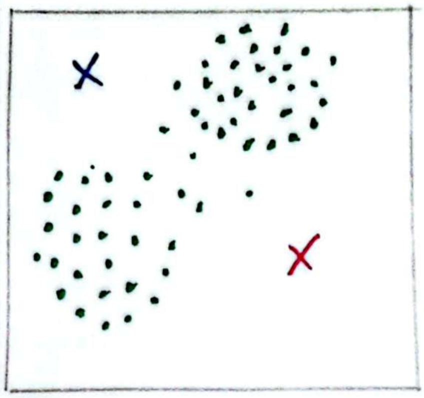
- The goal of clustering is to create groups of data points such that point in different clusters are dissimilar while points within a cluster are similar.
- With K-Mean Clustering, we want to cluster our data points into K groups.
- The output of the algorithm would be a seat of "~~labels~~/classes" assigning each data point to one of the K groups.

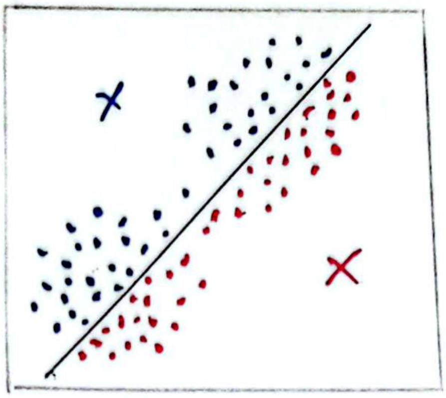
In K-Mean Clustering, the way these groups are defined by creating a centroid for each group. The centroids are like the heart of the cluster, they "capture" the points closest to them and add them to the cluster.

## Algorithm Steps

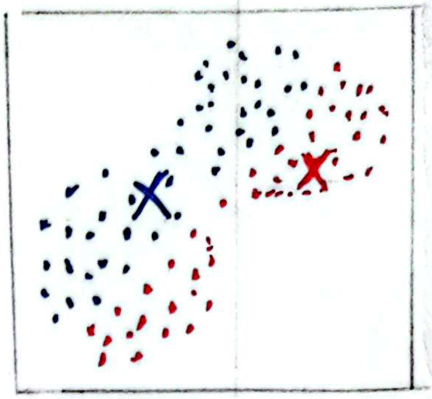
1. Define the K centroids → Randomly
2. Update cluster assignments → Assign each data point to one of the K clusters which is nearest.
3. Update Centroids → Move the centroid to the center of their clusters.

Repeat step 2 and 3 until convergence.

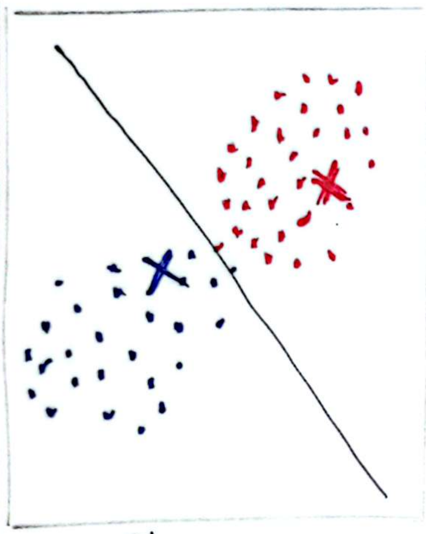




Step-2



Step-3



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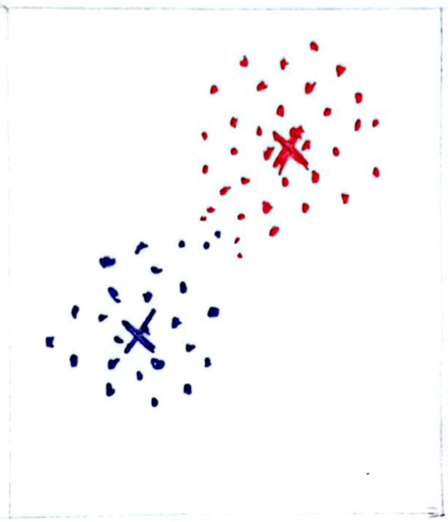
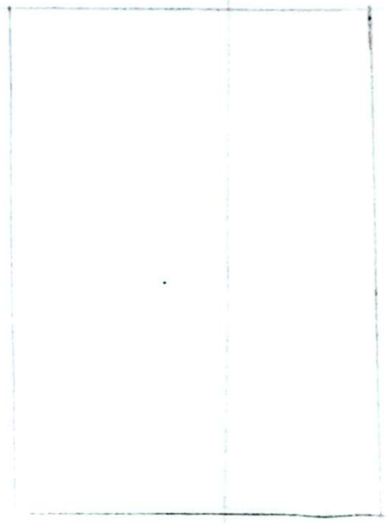
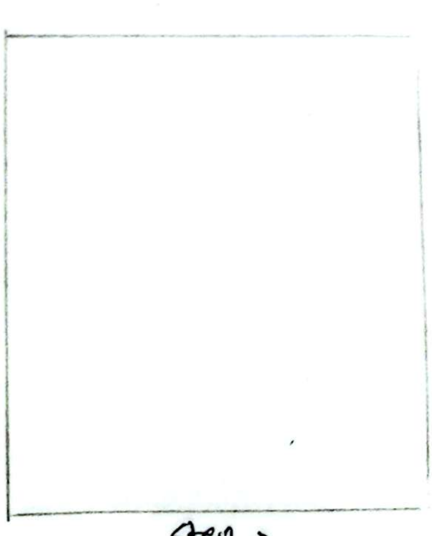


Step-3



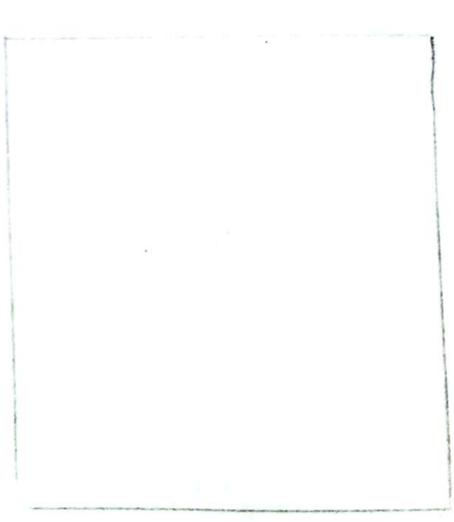
Step-2

Similarly



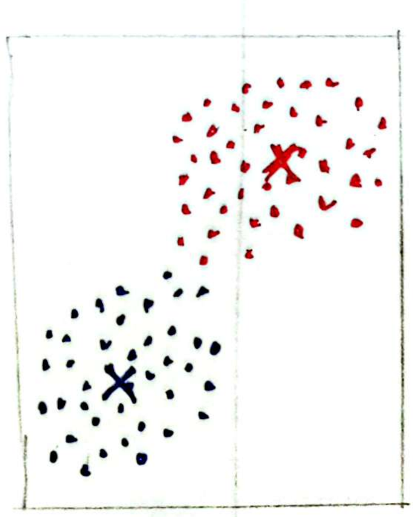
Step-3

Similarly

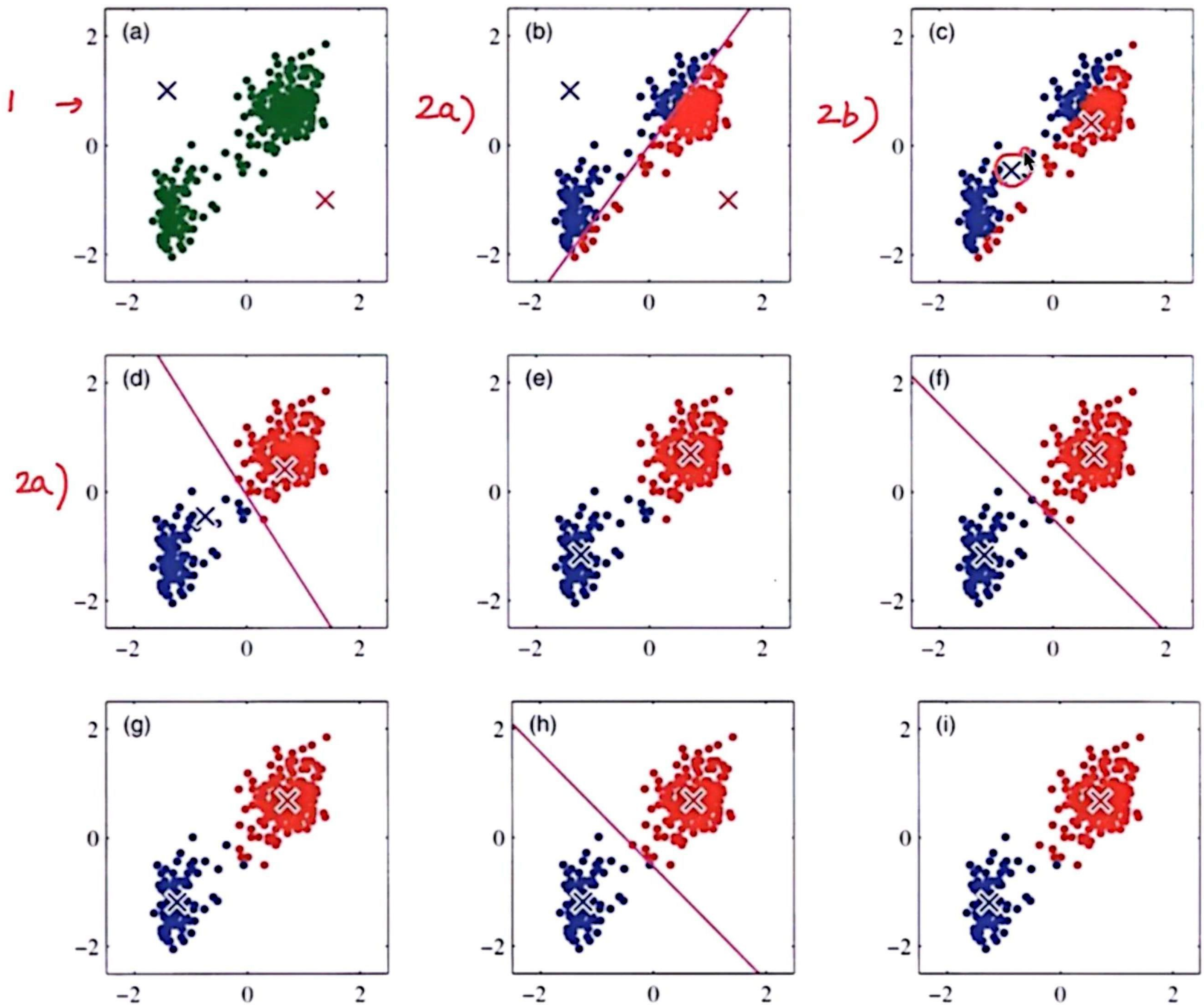


Step-2

Similarly



Step-3



1 →
(a)
2a)
(b)
2b)
(c)
2a)
(d)
(e)
(f)
(g)
(h)
(i)